

PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CANDIDATE
NAME

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CT
GROUP

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INDEX
NUMBER

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CHEMISTRY

9729/03

Paper 3 Free Response

19 September 2018

2 hours

Candidates answer on separate paper.

Additional Materials: Answer Paper
 Data Booklet
 Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CT group on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **one** question.

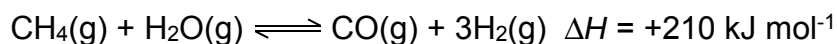
The use of an approved scientific calculator is expected, where appropriate. A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Section A

Answer all questions in this section.

- 1 (a) Steam reforming is an important industrial method for producing hydrogen and carbon monoxide, from hydrocarbon fuels. This is achieved in a processing device called a reformer which reacts steam at high temperature with the fuel, such as methane gas.



- (i) Write an expression of K_p for the reaction, indicating its units. [2]

$$K_p = \frac{(P_{\text{CO}})(P_{\text{H}_2})^3}{(P_{\text{CH}_4})(P_{\text{H}_2\text{O}})} \quad \text{atm}^2$$

- (ii) At temperature T , a mixture of gaseous methane, steam and carbon monoxide were introduced into an evacuated reactor.

The table below shows the initial pressures of the gases before they were added to the reactor.

Gas	Initial pressure / atm
methane, CH_4	1.50
steam, H_2O	2.50
carbon monoxide, CO	0.70

Given that the percentage dissociation of methane is 20% at equilibrium, find the K_p for the reaction. [2]

	$\text{CH}_4(\text{g})$	$+ \text{H}_2\text{O}(\text{g}) \rightleftharpoons$	$\text{CO}(\text{g})$	$+ 3\text{H}_2(\text{g})$
Initial P / atm	1.50	2.50	0.70	0
Change P / atm	-0.2(1.5) = -0.3	-0.3	+0.3	+0.9
Eqm P / atm	1.20	2.20	1.00	0.90

$$K_p = \frac{(P_{\text{CO}})(P_{\text{H}_2})^3}{(P_{\text{CH}_4})(P_{\text{H}_2\text{O}})}$$

$$K_p = \frac{(1.00)(0.90)^3}{(1.20)(2.20)} = 0.276 \text{ atm}^2$$

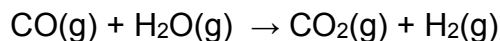
- (iii) A chemist decided to raise the temperature of the reactor. He suggested that this would result in a lower yield of the product.

Do you agree with the chemist? Explain your answer.

[2]

No. When the temperature is raised, the endothermic reaction is favoured since it absorbs energy. The position of equilibrium shifts to the right in favour of the forward reaction, yield of H_2 should increase.

In order to produce more hydrogen gas, the carbon monoxide gas that is produced from the steam reforming reaction can further react with steam. This is known as the “water-gas shift” reaction.



- (iv) Given that the reaction conditions were set at 500°C and 1.5 atm, determine the maximum volume of hydrogen gas that can be produced from 5kg of carbon monoxide gas.

State the assumption made in your calculation.

[3]

Amount of CO in 5kg = $(5 \times 1000)/28 = 178.6 \text{ mol}$ = Amount of H_2 gas

Amount of H_2 produced = 178.6 mol

$$pV = nRT$$

$$V = \frac{nRT}{p}$$

$$V = \frac{178.6 \times 8.31 \times (500 + 273)}{1.5 \times 1.01 \times 10^5}$$

$$= 7.55 \text{ m}^3$$

Assumption: Hydrogen gas is behaving like an ideal gas

- (v) Given the following,

enthalpy change of formation of CO(g) / kJ mol^{-1}	-99
enthalpy change of formation of $\text{H}_2\text{O(g)}$ / kJ mol^{-1}	-245
enthalpy change of formation of $\text{CO}_2\text{(g)}$ / kJ mol^{-1}	-394

With reference to the above data, calculate the enthalpy change of reaction of the “water-gas shift” reaction, ΔH_1 . [1]

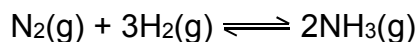
$$\text{CO(g)} + \text{H}_2\text{O(g)} \rightarrow \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$$

$$\Delta H_1 = \Delta H_f(\text{CO}_2\text{(g)}) - [\Delta H_f(\text{CO(g)}) + \Delta H_f(\text{H}_2\text{O(g)})]$$

$$= (-394) - (-99 + -245) = -50 \text{ kJ mol}^{-1}$$

- (b) The hydrogen gas obtained from the steam reforming process in part (a) can be reacted with nitrogen gas to produce ammonia gas.

This reaction is known as the Haber-Bosch Process, using iron as a catalyst.



- (i) Predict and explain the sign of ΔS for the Haber-Bosch reaction. [1]

ΔS is negative as there is a decrease in number of moles of gaseous particles (from 4 mol to 2 mol), hence less disorderliness.

- (ii) Using your answer in (b)(i), calculate the maximum temperature for the Haber-Bosch reaction to be spontaneous.

It is known that the numerical value of ΔS is $198 \text{ J K}^{-1} \text{ mol}^{-1}$, and that $\Delta H = -92.0 \text{ kJ mol}^{-1}$. [2]

$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ 0 &= (-92.0) - T(-198/1000) \\ T &= 465\text{K}\end{aligned}$$

- (c) The hydrogen gas obtained from steam reforming, and the ammonia gas obtained from the Haber-Bosch reaction, are the key reagents required for the reductive amination reaction.

Many amines in the pharmaceutical drugs are synthesised through this reaction.

The diagram, **Fig. 1.1**, shows the 3 stages involved in reductive amination.

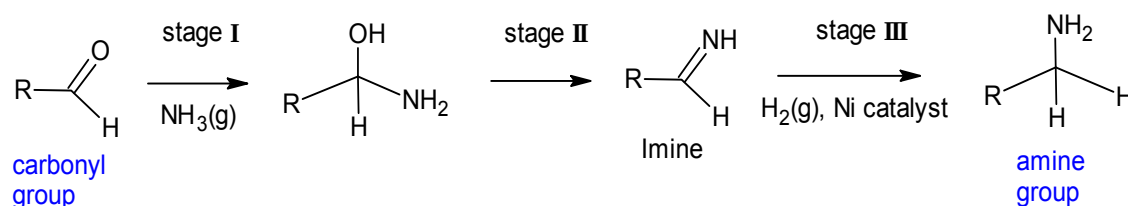


Fig. 1.1

During the reaction, a carbonyl group is converted into an amine group via an intermediate imine.

- (i) State the types of reaction occurring in stage I and stage II. [2]

stage I: Nucleophilic addition.
stage II: elimination

The reaction scheme below, **Fig.1.2**, shows the production of α -methylphenethylamine, which involves reductive amination in step 4.

Step 4 in **Fig. 1.2** comprises of the 3 stages of reductive amination in **Fig. 1.1**.

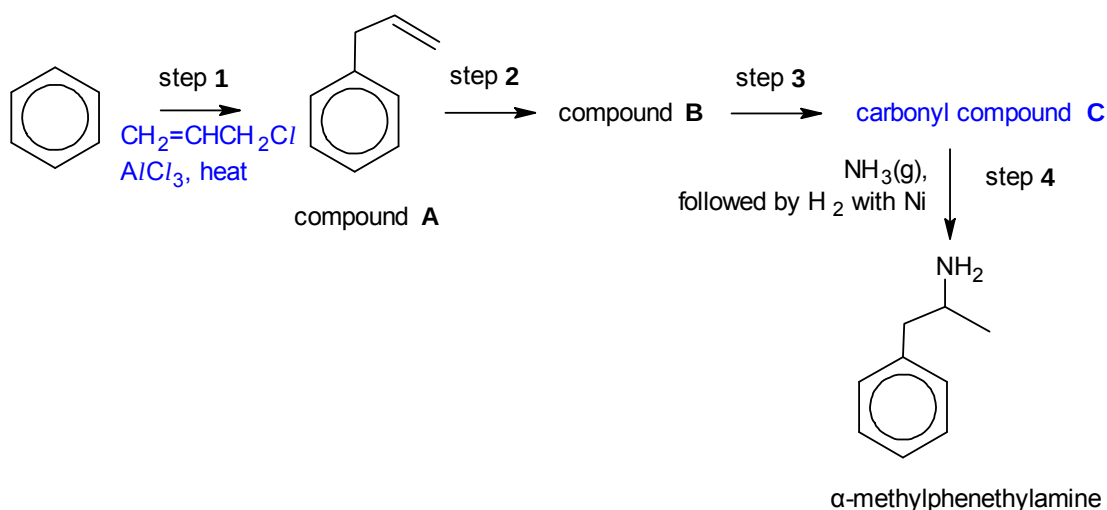


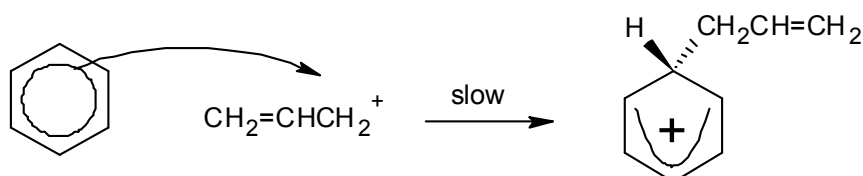
Fig.1.2

- (ii) In step 1, benzene undergoes a reaction with 3-chloroprop-1-ene with the aid of anhydrous aluminium chloride catalyst.

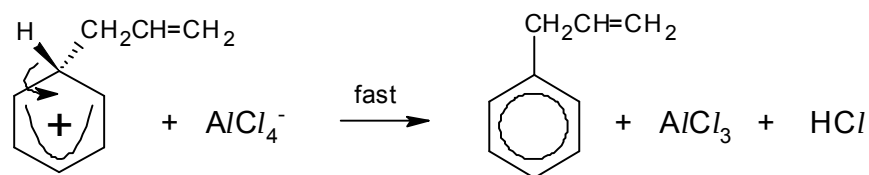
Name and describe the mechanism for step 1, showing clearly any relevant charges, dipoles or lone pairs of electrons that are important. [4]
 Electrophilic Substitution



Step 2:

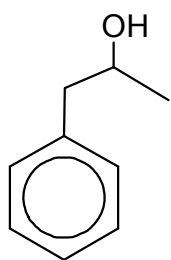


Step 3:

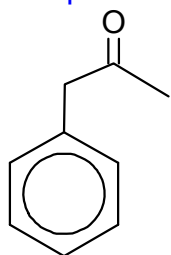


- (iii) Deduce and draw the structure for carbonyl compound **C**, and suggest the structure of compound **B**. [2]

compounds **B**:



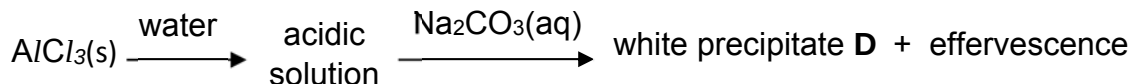
compounds C:



[Total: 21]

2 This question is on halogen-containing compounds.

(a) The following experiments are carried out starting from solid aluminium chloride.

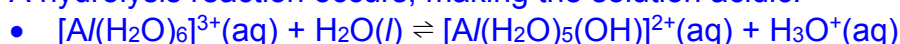


(i) Explain with the aid of balanced equations, why $\text{AlCl}_3(\text{s})$ dissolves to form an acidic solution. Suggest the pH of the acidic solution. [3]

$\text{AlCl}_3(\text{s})$ dissolves in water to form $[\text{Al}(\text{H}_2\text{O})_6]^{3+}(\text{aq})$



A hydrolysis reaction occurs, making the solution acidic.



• Since Al^{3+} has high charge density, it polarises and weakens the O-H bond of the water attached to it, causing the O-H bond to break to release H^+ (hydrolysis)

• pH=3

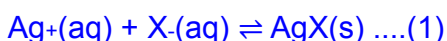
(ii) Identify precipitate D.

[1]



(b) When aqueous AgNO_3 is added to separate test-tubes containing aqueous KCl and KI , the various AgX ($\text{X} = \text{Cl}, \text{I}$) precipitates were observed.

With the aid of relevant equations, explain why a precipitate of AgCl is soluble in excess aqueous NH_3 but a precipitate of AgI is insoluble in excess aqueous NH_3 . [3]



When excess NH_3 is added (high $[\text{NH}_3]$), the position of equilibrium for (2) shifts to the right to form complex $[\text{Ag}(\text{NH}_3)_2]^+$ and $[\text{Ag}^+]$ decreases, resulting in eqm (1) to shift left & a decrease in ionic product of AgX .

For AgCl , IP decreases, ionic product becomes smaller than $K_{\text{sp}}(\text{AgCl})$ and the position of equilibrium (1) shifts to the left and AgCl dissolves completely.

For AgI , although IP decreases, ionic product is still higher than $K_{\text{sp}}(\text{AgI})$ (as $K_{\text{sp}}(\text{AgI})$ is very small) and thus there is no dissolution of AgI in excess aqueous ammonia.

(c) The standard redox potential of chlorate (V) ions is given below.



Predict what you expect to observe, when acidified potassium chlorate (V) is added separately to each of the following reagents.

State clearly if no reaction occurs. Write a balanced equation for any reaction that occurs. Explain your answer using $E^\ominus$ values from the *Data Booklet*.

(i) FeSO_4 (aq)

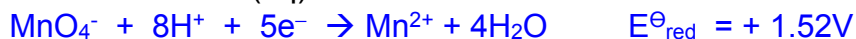
From the eqn given and highly positive $E^\ominus = 1.45 \text{ V}$, ClO_3^- is easily reduced.
For redox to occur, Fe^{2+} must be oxidised.



$E^\ominus_{\text{cell}} = +1.45 + (-0.77) = +0.68 \text{ V} > 0$ (3 s.f) reaction is energetically feasible

Solution turns yellow (from green).

(ii) Acidified KMnO_4 (aq)



MnO_4^- and ClO_3^- both can only be reduced – thus no redox possible and solution remains purple

[3]

(d) PCl_5 reacts with hydrazine, N_2H_4 , to give compound **E**, which has the following composition by mass: P, 20.5%; N, 9.2%; Cl, 70.3%. The molecule of **E** contains an N-N single bond.

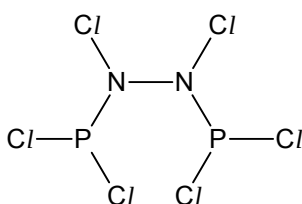
Calculate the empirical formula of **E**, and suggest its molecular formula and the structure of its molecule, assuming its constituent atoms show their usual valencies.

[2]

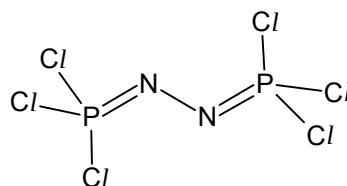
	P	N	Cl
Mass / g	20.5	9.2	70.3
Amt / mol	0.661	0.657	1.98
Mole ratio	1	1	3

Empirical formula = PNC_3

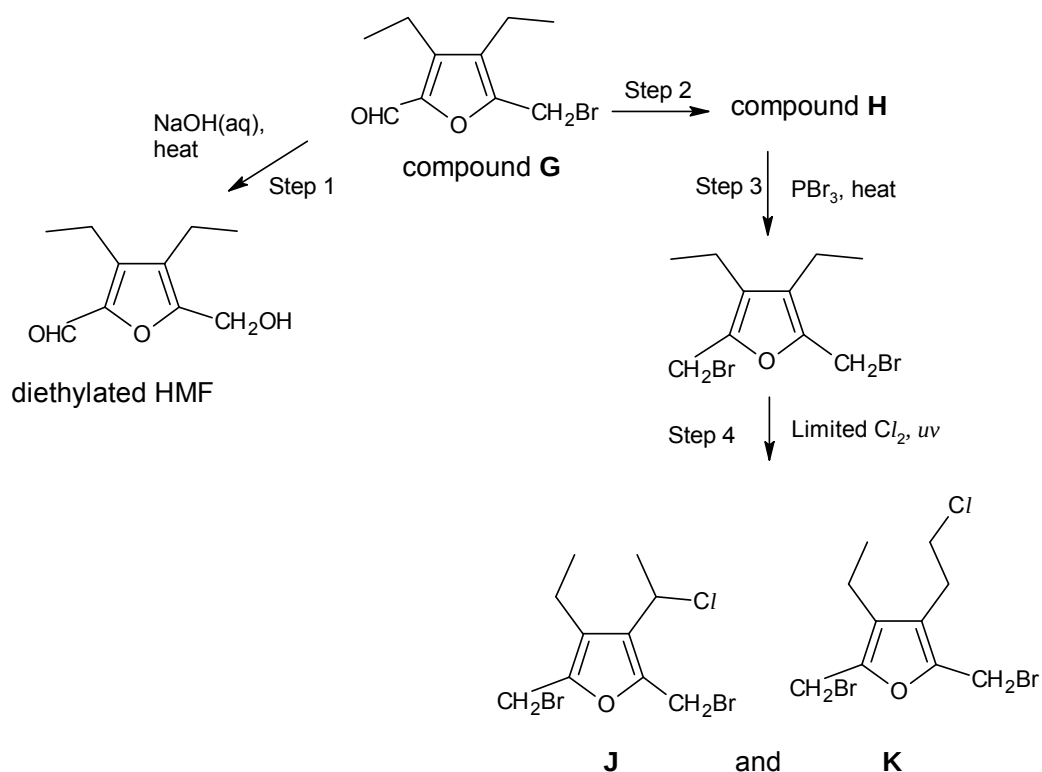
Since **E** contains one N-N single bond, **E** will contain 2 N atoms.
Molecular formula of **E** = $\text{P}_2\text{N}_2\text{Cl}_6$ and correct structures of **E**:



or



(e) Diethylated HMF is used to synthesize a variety of pharmaceuticals. The flow-scheme below shows the reactions involving diethylated HMF and compound **G**, its precursor. Assume the C-O-C ether group is inert.



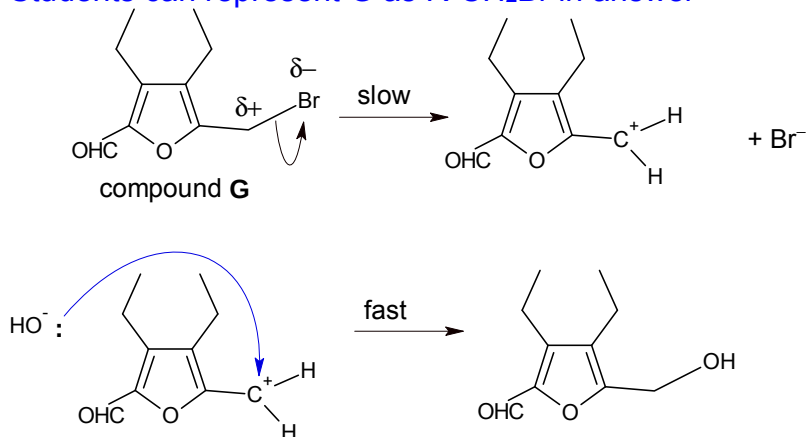
- (i) When the reaction in step 1 was investigated kinetically, it was found that its rate was independent of $[\text{OH}^-]$, but was first order with respect to [compound G].

Name and describe a mechanism for this reaction, showing curly arrows in your answer. [3]

You may represent compound G as $\text{R-CH}_2\text{Br}$.

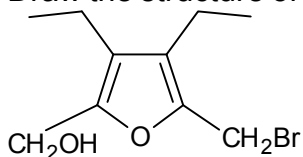
Nucleophilic substitution ($\text{S}_{\text{N}}1$)

Students can represent G as $\text{R-CH}_2\text{Br}$ in answer

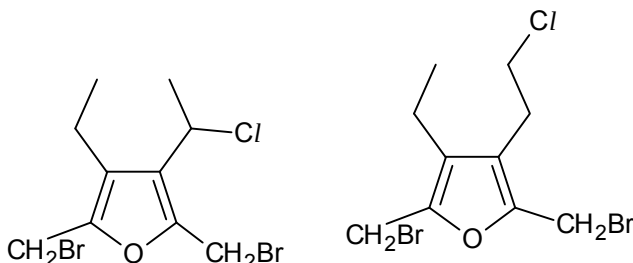


- (ii) Draw the structure of compound
- H**
- .

[1]



- (iii) When the mono-chlorination is carried out in step 4, two possible monochloroalkanes
- J**
- and
- K**
- can be formed.

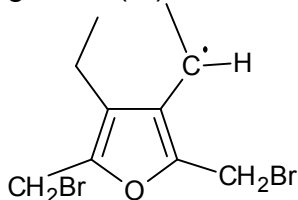
**J** and **K**Predict the relative proportions of **J** and **K** formed.

[1]

J : **K**
 4: :6
 OR 2: 3

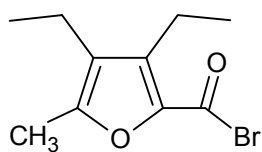
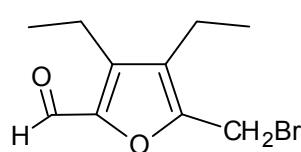
- (iv) When the monochlorinated products of step 4 were analysed, it is found that the mole ratio of
- J**
- :
- K**
- formed was about 97: 3.

Suggest a reason for the difference between this ratio and the one you gave in (iv). [1]



is exceptionally stable due to the overlap of p-orbital between the Carbon with unpaired electron (or resonance stabilised), and the p-orbitals of the adjacent alkene carbon atoms. This is an allylic radical. Being highly stable, it is formed in greater amount / faster and leads to major product.

- (v) Compounds
- M**
- and
- G**
- are both bromine-containing compounds that are isomers of each other.

Compound **M**Compound **G**

Compare the reactivities of **M** and **G** with water. Explain your answer.
[2]

- Rate/ ease of hydrolysis: $M > G$

M, an acyl bromide is most reactive; it undergoes hydrolysis most readily
Reason

- The additional strongly electronegative oxygen atom in acyl bromide M further increases the partial positive charge on carbon atom (of COBr) or make it more δ^+ positive compared to carbon atom (of C-Br) in G.

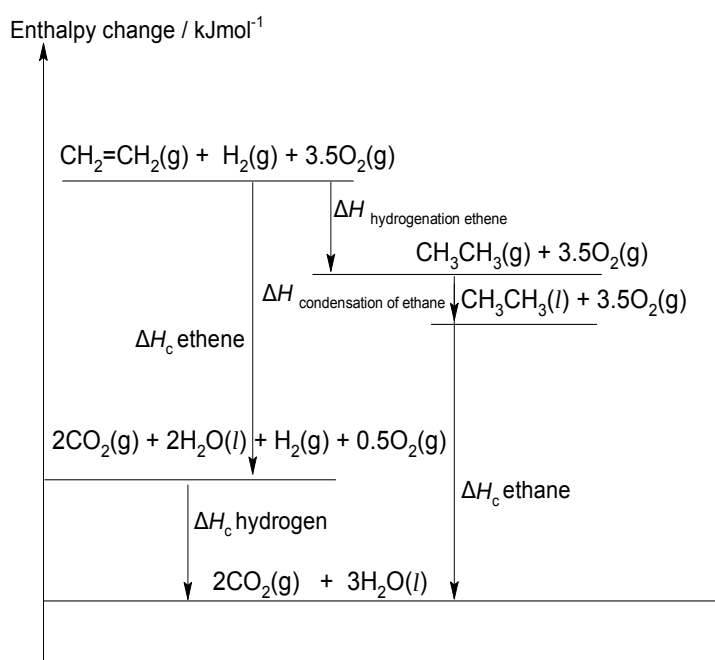
Thus the carbon atom in X is *more* readily attacked by nucleophile or X undergoes hydrolysis by water readily to give bromide ions.

[Total:20]

- 3 (a) (i) Use the following data to construct an energy level diagram and hence calculate the enthalpy change for the hydrogenation of ethene gas ($\text{CH}_2=\text{CH}_2$) to ethane gas (CH_3CH_3).

	$\Delta H / \text{kJ mol}^{-1}$
Enthalpy change of combustion of $\text{CH}_3\text{CH}_3(\text{l})$	-1545
Enthalpy change of condensation of $\text{CH}_3\text{CH}_3(\text{g})$	-15
Enthalpy change of combustion of $\text{H}_2(\text{g})$	-286
Enthalpy change of combustion of $\text{CH}_2=\text{CH}_2(\text{g})$	-1411

[4]



By Hess' Law,

$$\Delta H_{\text{hydrogenation}} + (-15) + (-1545) = (-1411) + (-286)$$

$$\Delta H_{\text{hydrogenation}} = -137 \text{ kJ mol}^{-1}$$

- (ii) Define the term bond energy. [1]

Bond energy is the energy absorbed to break one mole of a covalent bond between two atoms in the gaseous state.

- (iii) Using appropriate bond energy values from the *Data Booklet* alone, calculate another value for the enthalpy change of the reaction in (a)(i).

$$\Delta H = \sum \text{B.E.}(\text{reactants}) - \sum \text{B.E.}(\text{products})$$

$$= \text{B.E.}(\text{C}=\text{C}) + 4\text{B.E.}(\text{C}-\text{H}) + \text{B.E.}(\text{H}-\text{H}) - [6\text{B.E.}(\text{C}-\text{H}) + \text{B.E.}(\text{C}-\text{C})]$$

$$= (+610) + 4(+410) - [(+350) + 6(+410)] = -124 \text{ kJ mol}^{-1}$$

[2]

- (iv) Suggest a reason for the discrepancy between the enthalpy change for the hydrogenation of ethene gas to ethane gas you calculated in (a)(i) and (a)(iii).

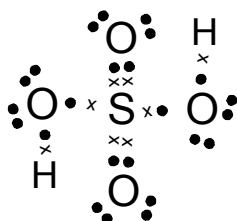
[1]

The bond energy values given in the Data Booklet are average values.

- (b) Hot sulfuric acid, H_2SO_4 is added to ethanol, to produce ethene.

Draw a 'dot-and-cross' diagram showing the electrons (outer shell only) in a H_2SO_4 molecule.

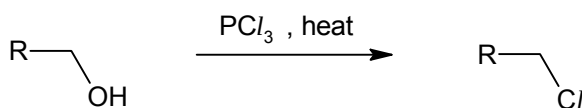
Use the VSEPR (valence shell electron pair repulsion) theory to predict the bond angle around sulfur in H_2SO_4 . [3]



According to VSEPR, order of repulsion is given by: Lone pair – lone pair electrons > Lone pair – bond pair electrons > Bond pair – bond pair electrons

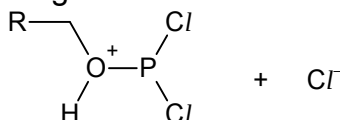
- The S atom has 4 bond pairs of electrons, no lone pair of electrons
- The bond angle is 109.5°

- (c) Phosphorous halides such as PX_3 can function as both Lewis acids or Lewis bases. PCl_3 is used to convert an alcohol to an alkyl chloride.



The mechanism of the above reaction involves two steps:

- Step 1: The nucleophilic alcohol attacks PCl_3 to form the following intermediate, along with a chloride ion:



- Step 2: (slow step) the chloride ion attacks the electrophilic carbon on the intermediate, at the same time the C–O bond breaks heterolytically.

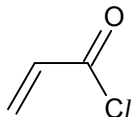
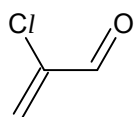
- (i) State and explain if PCl_3 acts as a Lewis acid or Lewis base in step 1 of the mechanism. [1]

PCl_3 is a Lewis acid because it accepted lone pair of electrons.

- (d) For the following pairs of compounds, describe one chemical test which would enable you to distinguish clearly between them.

State the observations for each compound.

State the type of reaction undergone for the compound that shows a positive test.



compound **N** compound **O**

Distinguishing test: Add Tollens' reagent, warm

Observations: Silver mirror for N, no silver mirror for O

Type of reaction: Oxidation

OR

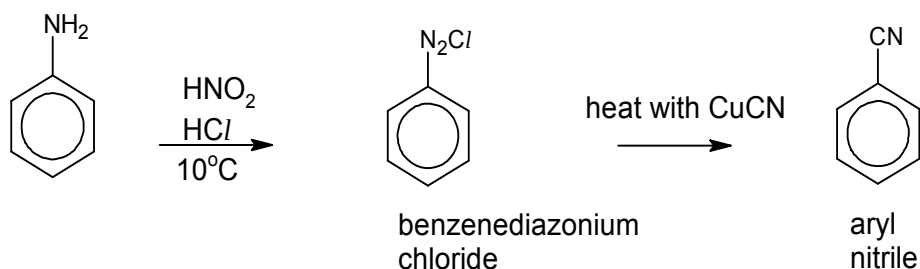
Distinguishing test: Add $\text{AgNO}_3(\text{aq})$, r.t.p

Observations: White ppt for O, no white ppt for N

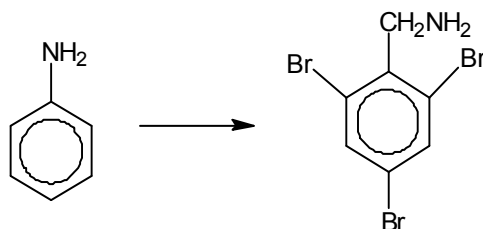
Type of reaction: Nucleophilic acyl substitution [3]

- (e) Phenylamine reacts with nitrous acid, HNO_2 , and hydrochloric acid at 10°C to form benzenediazonium chloride, $\text{C}_6\text{H}_5\text{N}_2\text{Cl}$.

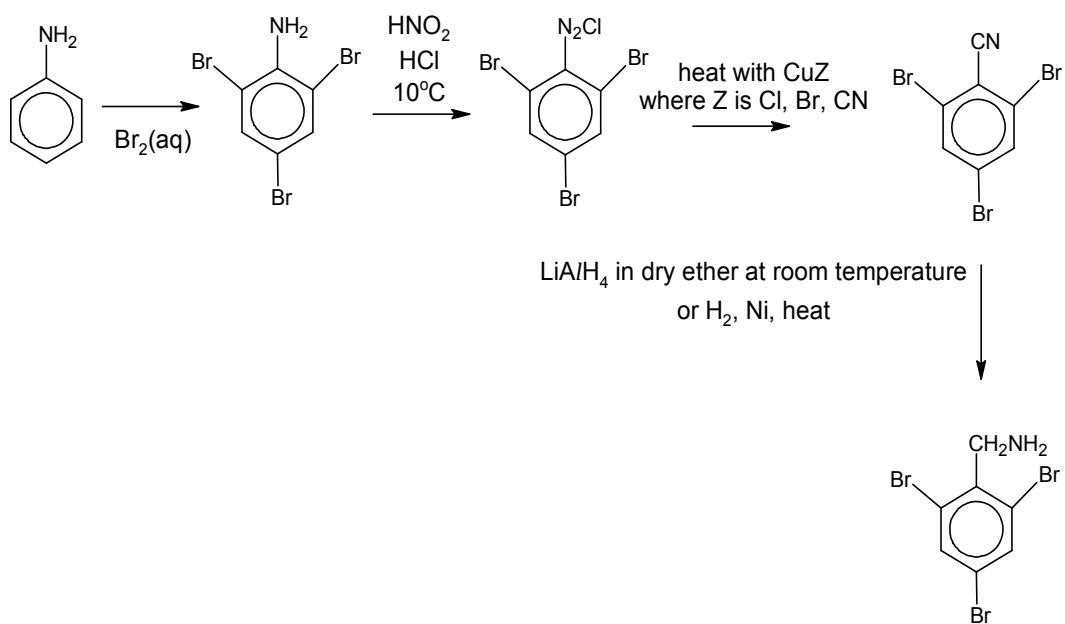
This is an important intermediate in the synthesis of aromatic compounds. It undergoes the Sandmeyer reaction, shown below, in the presence of a copper(I) catalyst to form aryl nitriles.



Suggest a four-step reaction sequence for the following conversion. You are to make use of the Sandmeyer reaction in one of the steps.



[4]



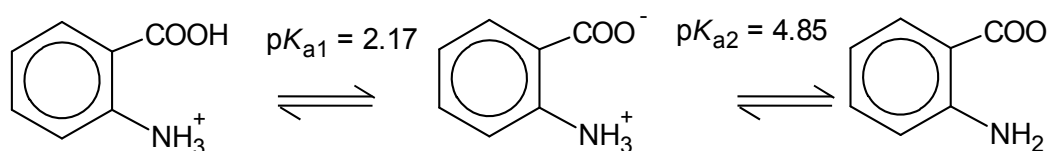
[Total:19]

Section B

Answer **one** question from this section.

- 4 (a) Vitamin L1, also known as 2-aminobenzoic acid, can be obtained from bovine liver, and its deficiency has been associated with lactation issues in females.

The diagram below shows how the protonated form of Vitamin L1 acts as a dibasic acid, which dissociates in two stages:



Protonated form
of Vitamin L1

- (i) Calculate the pH of 10 cm³ of 0.100 mol dm⁻³ of the protonated form of Vitamin L1. Ignore the effect of pK_{a2} on the pH of the sample. [1]

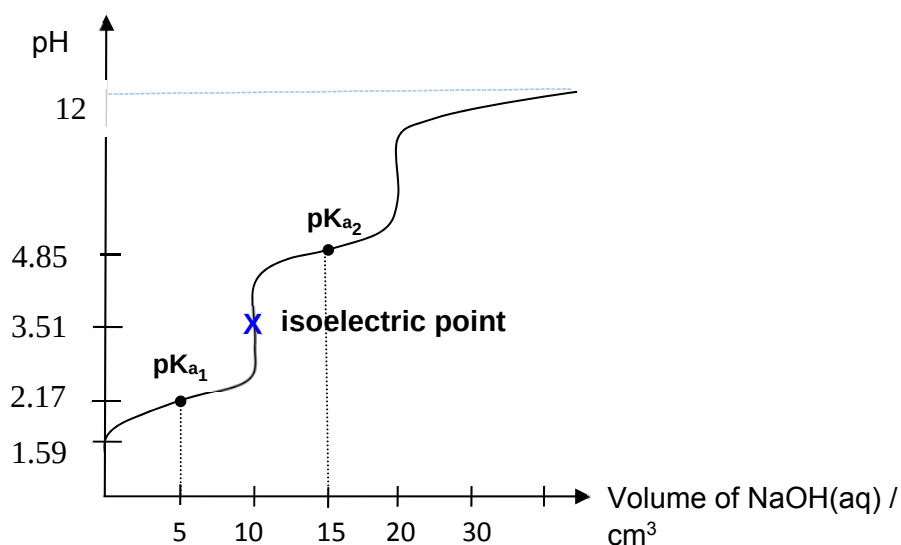
Given $pK_{a1} = 2.17$
 $K_{a1} = 10^{-2.17}$
 $K_{a1} \approx [H^+]^2 / (0.100)$
 $[H^+] = 0.02600 \text{ mol dm}^{-3}$
 $pH = 1.59$

- (ii) A sample containing 10 cm³ of 0.100 mol dm⁻³ of the protonated form of Vitamin L1 was titrated with 30 cm³ of 0.100 mol dm⁻³ of NaOH.

Sketch clearly the titration curve obtained, which should include the following key points:

- initial pH;
- points where pK_{a1} and pK_{a2} occur;
- isoelectric point of Vitamin L1 and its estimated value, given that:
isoelectric point = (pK_{a1} + pK_{a2}) / 2

[4]



Correct shape of graph with correct axes

Correct locations of pK_{a1} and pK_{a2}

Correct initial pH AND correct equivalence volumes

Credit for show pH tapering off at about 12-13 at about 30 cm^3

value of pH at isoelectric point

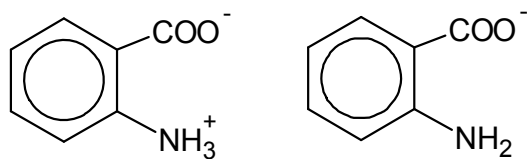
isoelectric point = $(pK_{a1} + pK_{a2}) / 2 = (2.17 + 4.85) / 2 = 3.51$

(iii) Define what is meant by a buffer. [1]

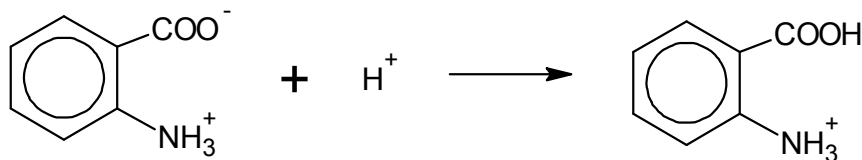
A buffer resists change to pH when small amount of acid or base is added to it and upon dilution.

(iv) Draw the structures of the two chemical species present in a solution of Vitamin L1 at pH 5.00.

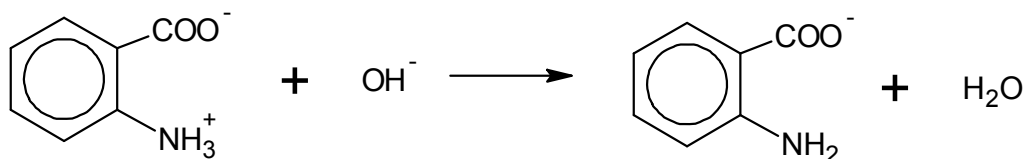
Hence, show by means of balanced equations, how this solution regulates the acidity on addition of H^+ ions and OH^- ions. [3]



When a small amount of acid is added,



When a small amount of base is added,



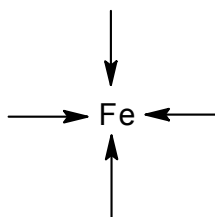
- (b) Iron ions can form transition metal complexes with the basic form of the amino acid, alanine. The formula of the complex formed is $\text{Fe}(\text{CH}_3\text{CH}(\text{NH}_2)\text{COO})_2$.

- (i) State the oxidation state of Fe in the transition complex. [1]

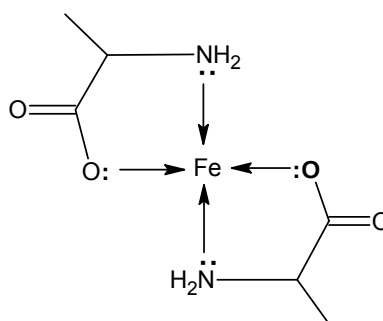
+2

Since the charge of each $\text{CH}_3\text{CH}_2(\text{NH}_2)\text{COO}^-$ is -1, iron has to be +2 for the complex to be electrically neutral as given.

- (ii) The diagram below shows the incomplete structure of the complex. Copy the diagram and draw the complete structure of this complex.

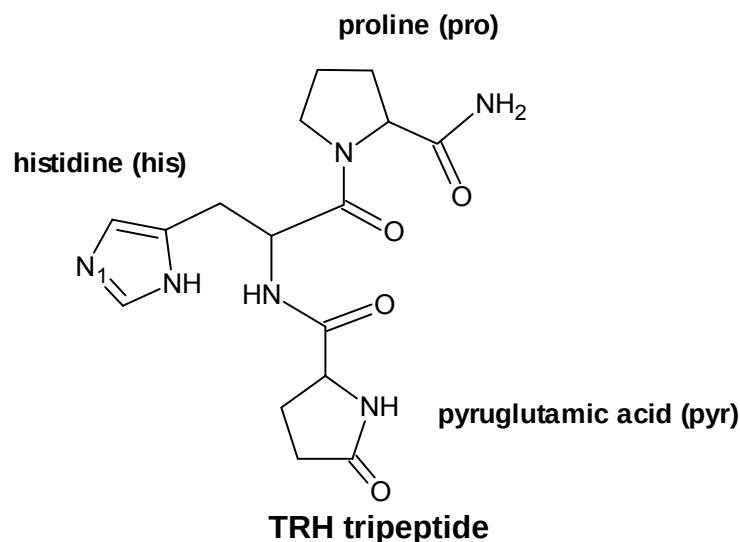


[1]



- (c) Thyrotropin-releasing hormone (TRH) is a hormone that is produced by the hypothalamus in the brain. TRH has been used clinically for the treatment of the disturbance of consciousness in humans.

TRH is a tripeptide that is made of three amino acids proline, histidine and pyruglutamic acid, as shown in the diagram below.



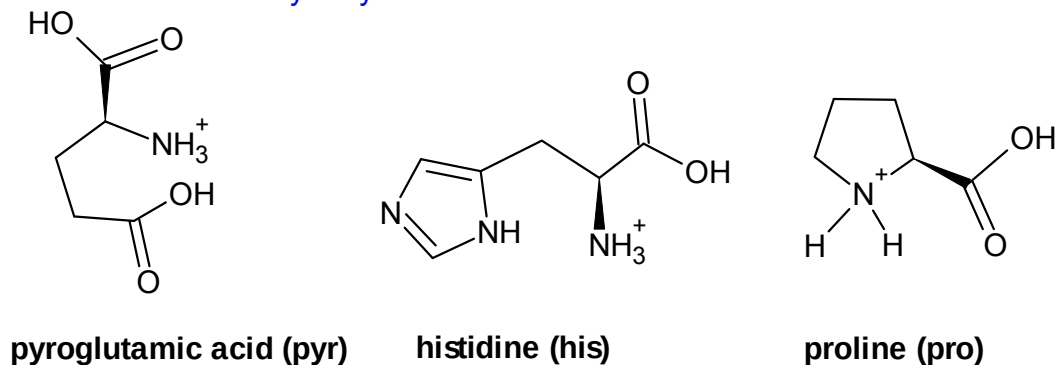
In the questions below, assume that the side chain for histidine (his) is inert.

- (i) State the reagents and conditions for TRH to undergo hydrolysis. [1]

$\text{H}_2\text{SO}_4(\text{aq})$, heat
OR
 $\text{NaOH}(\text{aq})$, heat

- (ii) Draw the structures of the amino acids formed when TRH undergoes hydrolysis using the reagents and conditions stated in (c)(i). [3]

Products of acidic hydrolysis:



**accept products of alkaline hydrolysis, if the candidate indicated that in (c)(i).*

- (iii) Explain why amino acids exist as solids with high melting points. [1]

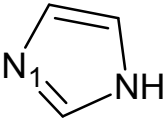
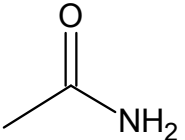
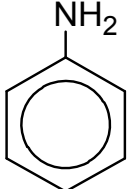
Amino acids can exist as zwitterions, with strong ionic bonds/electrostatic forces of attractions between oppositely charged ions.

- (iv) State the hybridisation state of nitrogen atom labeled N_1 in TRH. [1]

nitrogen atom 1: sp^2

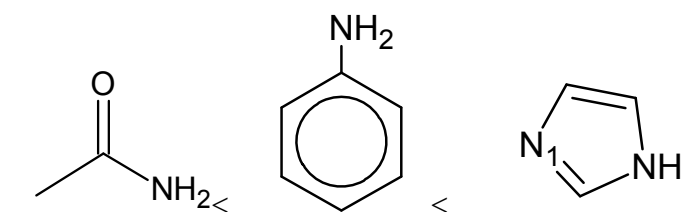
- (v) The amino acid histidine has a cyclic imidazole side-chain.

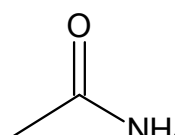
The table below compares the pK_b value of N_1 nitrogen atom of imidazole ring, with the pK_b values of ethanamide and phenylamine.

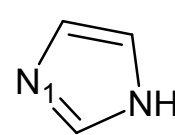
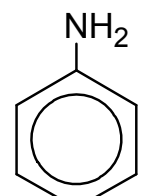
Compound name	Structure	pK_b
imidazole		8.8 (pK_b of N_1)
ethanamide		14.5
phenylamine		9.4

Using the data provided, list the 3 compounds in increasing basicity, and suggest explanations for their relative basicities. [3]

Basicity increases from:

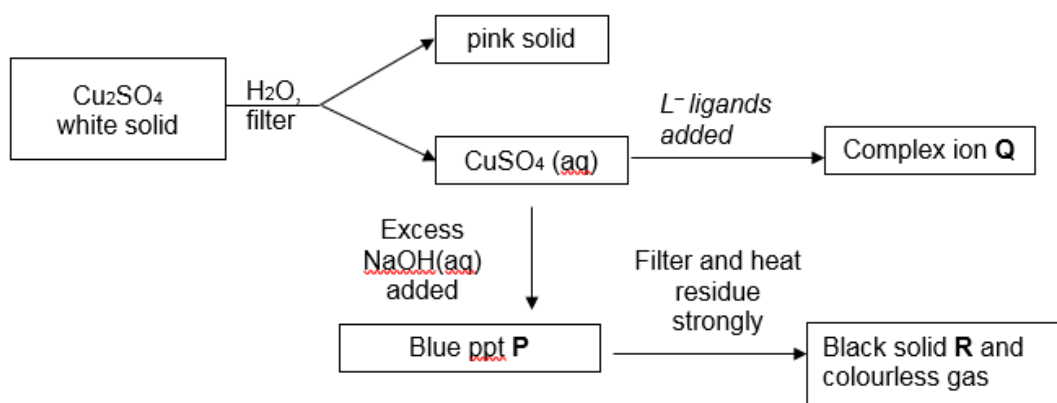


 is neutral as the lone pair of electrons on N is delocalised into $C=O$. Hence, the lone pair of electrons on the N atom is not available for protonation.

 is more basic than  as the p orbital of nitrogen lies perpendicular to the plane of the $C=N$ bond, hence, the lone pair of electrons are not delocalized and are readily available for protonation, compared to the lone pair of electrons on N atom in phenylamine which is delocalized into the benzene ring and not so readily available.

[Total: 20]

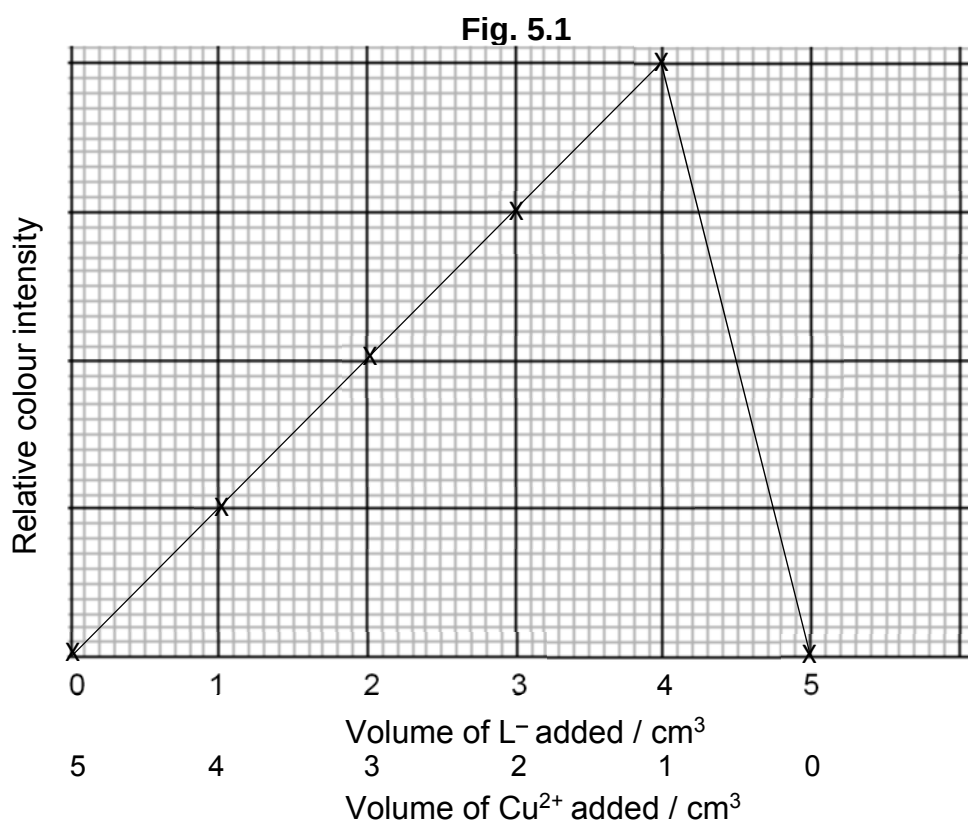
- 5 (a) The scheme below shows the reactions of different compounds of copper.



- (a) (i) Suggest the formulae of blue ppt **P** and black solid **R**. [2]
 P: $\text{Cu}(\text{OH})_2$
 R: CuO
- (ii) Write an equation to show how the pink solid and CuSO_4 are obtained when Cu_2SO_4 is added to water. [1]
 $\text{Cu}_2\text{SO}_4 (\text{s}) \rightarrow \text{CuSO}_4(\text{aq}) + \text{Cu}(\text{s})$

To determine the stoichiometry of the green complex ion **Q** formed between aqueous copper (II) sulfate and L^- ligands, the colour intensities of solutions containing different proportions of the ligand, L^- , and Cu^{2+} ion were measured.

The intensity of the green colour is directly proportional to the concentration of the complex ion. The higher the concentration of the complex, the higher is the colour intensity. In one such experiment, the results obtained for different volume mixtures of $1.0 \times 10^{-3} \text{ mol dm}^{-3} Cu^{2+}(aq)$ and $1.0 \times 10^{-3} \text{ mol dm}^{-3} L^-(aq)$ were plotted below.



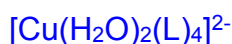
- (iii) State the type of reaction that occurs when complex ion **Q** is formed from aqueous $CuSO_4$. [1]

Ligand exchange

- (iv) Based on the plotted graph in **Fig. 5.1**, determine the mole ratio of $Cu^{2+}(aq)$ to $L^-(aq)$ in the complex **Q**.

Given that complex **Q** has an octahedral shape and it retains some water molecules as ligands, suggest a formula for the complex **Q**, including its overall charge. [2]

Mole ratio of $Cu^{2+}(aq)$ to $L^-(aq)$ in the complex **Q**: 1:4



- (b) Explain why complexes of Cu(II) are coloured whereas complexes of Cu(I) are colourless. [4]

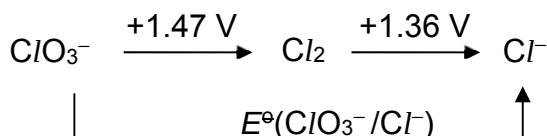
Cu(I) has fully filled 3 d-orbitals ($3d^{10}$). Hence, d-to-d electron transition cannot occur, and no visible light is absorbed, which accounts for it being colourless and Cu(II) has partially filled 3d-orbitals

- In presence of ligands,
- the originally degenerate 3d orbitals split into 2 sets of different energy levels, separated
- by a small energy gap (ΔE).
- ΔE corresponds to the energy of visible light.
- An electron from the lower energy d orbital absorbs a specific wavelength of visible light and
- is promoted to a half-filled, higher energy 3d orbital. d-to-d transition can take place and
- colour observed is complementary to the colour absorbed.

- (c) Chlorine can exist in different oxidation states.

Half-equation	E° / V
$2ClO_3^- + 12H^+ + 10e^- \rightleftharpoons Cl_2 + 6H_2O$	+1.47
$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$	+1.36

A Latimer diagram shown below, summarises the standard electrode potential, E° of different chlorine-containing species in acidic medium. The most highly oxidised form of the element is on the left, with successively lower oxidation states to the right. The electrode potentials are shown on the arrows.



The standard electrode potential of converting ClO_3^- to Cl^- , $E^\circ(ClO_3^- / Cl^-)$ is **NOT** the summation of +1.47 V and +1.36 V.

Determine the number of electrons involved in the conversion of 1 mole of ClO_3^- to 1 mole of Cl^- .

Hence, given that ΔG° for the conversion of ClO_3^- to Cl^- is -150 kJ mol^{-1} , calculate $E^\circ(ClO_3^- / Cl^-)$. [2]

No of electrons involved = 6 (+5 to -1)

$$\Delta G^\circ = -nFE^\circ$$

$$-150\,000 = -(6)(96500) E^\circ$$

$$E^\circ = +0.259 \text{ V}$$

(d) Use of the Data Booklet is relevant to this question.

X, $C_5H_6O_3$, is soluble in dilute NaOH. **X** gives an orange precipitate with 2,4-dinitrophenylhydrazine reagent, and a yellow precipitate with warm alkaline aqueous iodine. **X** decolourises aqueous bromine.

When it is reacted with a methanolic solution of $NaBH_4$ it gives **Y**, $C_5H_8O_3$. Catalytic hydrogenation of **X** gives **Z**, $C_5H_{10}O_3$. When **Z** is reacted with alkaline aqueous iodine, it gives $^-O_2C(CH_2)_2CO_2^-$.

Oxidation of 1 mole of **X** with hot acidified potassium manganate(VII) gives 1 mole of CH_3COCO_2H and 2 moles of CO_2 gas.

Additional optional information: X, Y and Z have a proton (1H) chemical shift value of 13.0 ppm. (Refer to Data Booklet Section 6: Typical proton (1H) chemical shift values (δ))

Suggest structures for **X**, **Y** and **Z**. Explain your reasoning clearly. [8]

- **X** is a carboxylic acid which undergoes acid base with NaOH.
- **X** is a ketone or aldehyde compound as it undergoes condensation reaction with 2,4-dinitrophenylhydrazine to form an orange ppt.
- **X** undergoes oxidation with warm alkaline aqueous iodine to form a yellow ppt, CHI_3 . **X** has CH_3CO- structure.
- **X** is an alkene as it undergoes electrophilic addition with Br_2 .
- Ketone or aldehyde group in **X** is reduced by $NaBH_4$ to give **Y** as no. of H atoms increases by 2.
- Both alkene and ketone group in **X** are reduced by hydrogen to give **Z** as no. of H atoms increases by 4.
- **Z** contains $CH_3CH(OH)-$ structure (not CH_3CO-) as it undergoes oxidation by alkaline aqueous iodine to give $^-O_2C(CH_2)_2CO_2^-$.

X is $CH_3COCH=CHCO_2H$

Y is $CH_3CH(OH)CH=CHCO_2H$

Z is $CH_3CH(OH)CH_2CH_2CO_2H$

[Total: 20]

End of paper